World Geospatial Industry Council

Bringing Geospatial Context to the Metaverse:

Considerations for the Next Steps



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Contents

1. Executive Summary **4**

2. Methodology and Scope **6**

3. Introduction **7**

4. Definitions of the Metaverse **10**

5. Building Geospatial Capabilities for the Metaverse **12**

5.1 Strategic Capabilities to Develop Geospatial Ecosystem in the Metaverse 12

5.2 Next Steps for Embedding Geospatial Capabilities into the Metaverse 16

6. Appendices **25**

Appendix 1 25

Examples of Early Implementation of the Metaverse 25

Appendix 2 26

Definitions of the Metaverse in Industry and Academia 26

Appendix 3 32

Esri 33

Hexagon AB 35

Oracle 36

Schneider Digital 38

TomTom 39

Trimble 41

Appendix 4 42

Overture Maps Foundation (OMF) 42

1. Executive Summary

The metaverse, a term coined by Neal Stephenson in his 1992 science-fiction novel *Snow Crash*, has been the subject of intense media interest since 2021. It has become a popular topic of discussion across a broad spectrum of society, with many commentators discussing the prospects of it mediating a significant part of our lives.

Although intense debate is happening on the exact parameters of the metaverse definition, most stakeholders agree that it is an interactive, immersive, multipurpose, multi-player, persistent, and online urban environment and has been used in gaming, academic research, and literary circles for nearly three decades. The World Geospatial Industry Council (WGIC) has formally adopted the following etymology based on what it believes to be the primary motivation behind the metaverse.

meta: *A prefix meaning beyond, transcending, or more comprehensive, and in this context, like metadata, is self-referential.*

universe: *The universe is everything. It includes all of space and all the matter and energy that space contains.*

metaverse: *A digital, and therefore, virtual, representation of the universe.*

This policy report brings geospatial context to the metaverse and provides specific considerations for the next steps towards expanding the value proposition of geospatial technology. The report examines the opportunities and risks for the geospatial industry in the burgeoning commercial metaverse market, estimated to be several trillion dollars in the next two to three decades. It looks at the definitions of the metaverse and how it relates to concepts such as digital twins, cyber geography, and virtual reality as we know them in the geospatial ecosystem. It presents a strategic outlook to develop a geospatial ecosystem in the metaverse, including online presence across multiple devices, data and technology for real-time multi-player productivity scenarios, real-time interoperability on a massive scale, user control of content and creation, and the provision of automation and performance using artificial intelligence (AI)/machine learning (ML).

The report highlights the significance of embedding geospatial capabilities into the metaverse and identifies four priorities for the next steps.

- ✔ **Priority 1** is understanding and navigating societal perceptions, as the metaverse will influence how we interact with each other and the world around us. These perceptions vary across age groups and geographies.
- ✔ **Priority 2** is to set up stakeholder engagement and collaboration to ensure that the geospatial industry is fully involved in developing the metaverse. These stakeholders will range from like-minded industry consortia to cautious policymakers in government departments and knowledge awareness forums in emerging economies.
- ✔ **Priority 3** is to investigate the complexities of multifaceted technological interoperability and data governance, as the metaverse will require interoperability across technologies and data sources.
- ✔ **Priority 4** is to create economic opportunities and market readiness markers to ensure that the geospatial industry can fully participate in the commercial opportunities presented by the metaverse value chain.

The report lists examples of early implementations of the metaverse and capabilities within the WGIC membership towards bringing the geospatial ecosystem to the metaverse.

The metaverse represents a significant opportunity for the geospatial industry to develop new products and services, generate new revenue streams, and create new jobs. However, it also presents several risks the geospatial ecosystem must address to ensure the ethical and responsible use of geospatial data in the metaverse. This report provides valuable insights for policymakers, practitioners, and researchers in the geospatial industry as they navigate the opportunities and challenges the metaverse presents.



2. Methodology and Scope

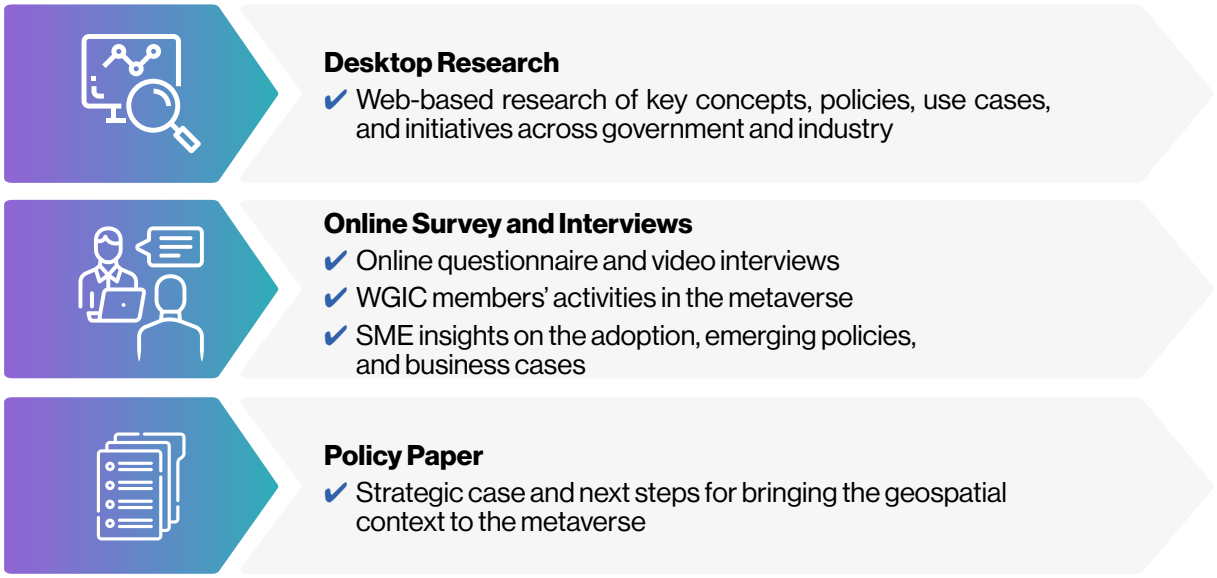
This paper proposes considerations for the geospatial stakeholders to reflect on and advance a broader view of the metaverse as a virtual representation of our physical universe.

The observations are based on user surveys and meta-analyses of articles published in academic journals and those published online by an ever-growing number of commentators and industry advisors. Specifically, the report provides:

- ✔ An understanding of the current state of engagement by the WGIC members and the geospatial industry.
- ✔ Pointers to global initiatives by industry bodies and public authorities towards promoting standardization, regulation, and investments in the development of the metaverse.

The study methodology involves three parts, as shown in **Figure 1**.

Figure 1: Three part-methodology to study the link between the geospatial ecosystem and the metaverse



Desktop research extends to academic journals, industry reports, media articles, and governmental publications. An in-depth online survey[1] and subsequent video interviews were conducted to gain the views of stakeholders. Over 30 global geospatial ecosystem subject-matter experts (SMEs) in areas ranging from WGIC member organizations, academia, government departments, non-governmental organizations, and think-tanks participated in detailed interviews. WGIC ran information campaigns promoting the survey on social media and at conferences.

It is important to note that metaverse is a vibrant and emerging field. Therefore, the questions and opportunities are continuously being thought of at the time of writing this report.

1 *https://forms.gle/TGGn8kwCncK9kR8j6*

3. Introduction

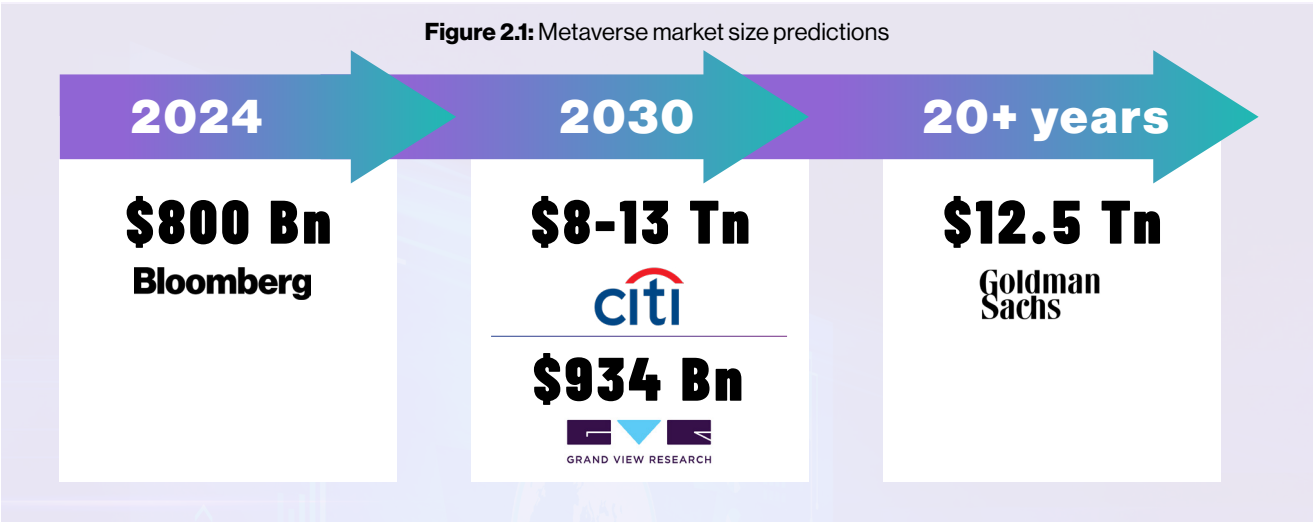
The metaverse has been the subject of intense media interest recently, with commentators discussing the prospects of it mediating significant aspects of our lives. But what is the metaverse, exactly?

And how does it relate to concepts such as spatial digital twins, cyber geography, and virtual reality (VR) as we know them in the geospatial ecosystem? What are the opportunities and risks for the geospatial industry in the burgeoning metaverse market?

The term "**metaverse**" was coined by Neal Stephenson in his 1992 science fiction novel *Snow Crash*[2] to describe a virtual reality experience used by its characters. The novel presents the metaverse as an interactive, immersive (3D), multipurpose, multi-player, and online urban environment. The novel's characters use the metaverse as a highly functional parallel realm of digital existence, with some even choosing to be connected to it continuously using head-mounted displays (HMDs), living double lives.

The metaverse was used in gaming, academic research, and literary circles for nearly three decades. However, in late 2021, Meta (previously Facebook) brought the term into the global spotlight, announcing their plans to become a metaverse company and making Neal Stephenson's concept a reality. To show their commitment, they invested heavily in VR products. To showcase the practical applications[3] of VR beyond gaming, Sir Nick Clegg, President of Global Affairs, Meta, participated in a VR-based interview with *Financial Times*[4]. This interview and the related promotions by Meta have led to a significant increase in market size predictions (**Figure 2.1**). New and emerging metaverse applications, such as financial trading, retail, health and virtual and immersive education (**Figure 2.2**), have started gaining traction.

Figure 2.1: Metaverse market size predictions



2 https://en.wikipedia.org/wiki/Snow_Crash
3 https://www.meta.com/en-gb/blog/quest/connect-future-of-work-vr/
4 https://www.ft.com/content/913c0cc2-9498-4d25-b2e8-794e24702e0b

Figure 2.2: Use cases of metaverse



Productivity & Collaboration
Digital twins of industries, smart cities, urban infrastructures



Finance
Blockchain and trading of cryptocurrencies, NFTs



Gaming
Multi-player virtual worlds



Health
Digital twins of humans for new medicine design; remote monitoring of patients



Retail and Entertainment
Immersive experience of purchasing fashion, homes, and music concerts



Immersive Learning
Virtual classrooms for students in inaccessible areas

Over 160 companies worldwide are actively involved in creating content and platform components for the metaverse at present[5], even as Gartner predicted the full maturity of the metaverse technology is up to ten years away (**Figure 3**). Nevertheless, significant recent investments pouring into the construction of entertainment, retail, and industrial metaverse indicate the level of anticipation and potential of the technology[6 7]. Several leading software and hardware players, such as Meta, NVIDIA, Unity, Microsoft, and Google, are actively involved in creating metaverse platforms. See **Appendix 1** for a list of frequently mentioned metaverse candidates.

The geospatial industry and academic communities are taking an interest in harnessing the potential of the metaverse because it provides an opportunity to develop holistic geospatial systems that can perform complex interactions (e.g., simulations for weather patterns using data from digital twins of the ocean, atmosphere, urban environment, geo-demography and insurance) between various geospatial data and social, economic, political, financial, and business systems. These interactions can range from technology-based systems such as Digital Earth[8] to the work of organizations such as Resilience Brokers[9], where technology and people are involved in environmental policymaking.

Geospatial data and technology are crucial for the success of the metaverse. Two specific facts support this belief:

- ✔ Significant similarities exist between the technologies and concepts utilized in the metaverse and geospatial fields. Specifically, VR worlds and associated technologies such as augmented/mixed/extended reality (AR/MR/XR)[10] have been extensively used in practical applications within the geospatial domain, including military

5 *https://metaverseinsider.tech/2022/08/08/7-of-the-worlds-top-metaverse-companies-in-2022/*
6 *https://www.bloomberg.com/news/articles/2023-07-25/japan-s-fumio-kishida-hints-at-metaverse-project-from-big-firm*
7 *https://press.siemens.com/global/en/pressrelease/siemens-invest-eu1-billion-germany-and-create-blueprint-industrial-Metaverse-nuremberg*
8 *http://www.digitalearth-isde.org/*
9 *https://resiliencebrokers.org/*
10 *https://en.wikipedia.org/wiki/Augmented_reality; https://en.wikipedia.org/wiki/Mixed_reality; https://en.wikipedia.org/wiki/Extended_reality*

simulations (e.g., simulations such as urban combat and flights), AEC design[11], environmental studies[12], urban planning[13] (visualization, collaborative decision making[14]), and education[15].

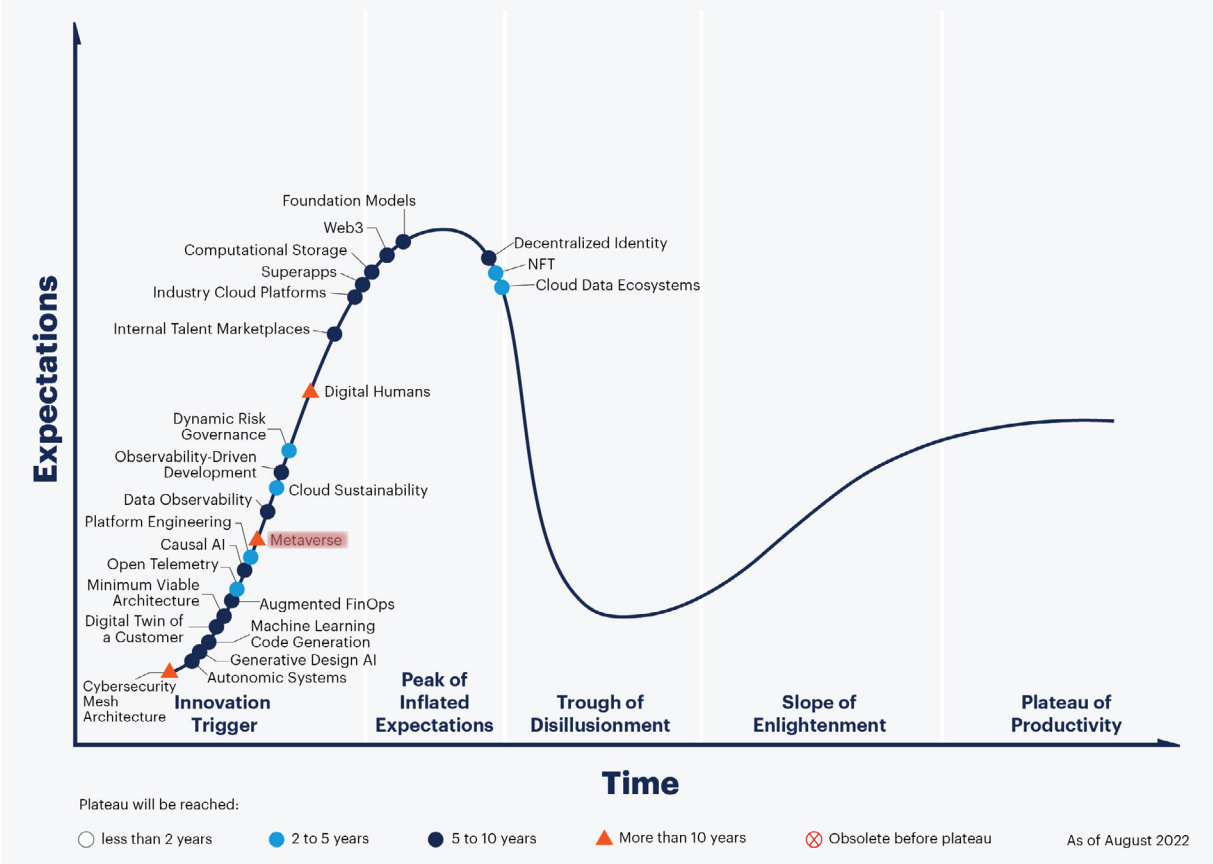


Figure 3: Gartner Hype Cycle for Emerging Technologies 2022 (Modified from Gartner Report, August 2022)

✔ Significant research has been conducted on accurately modeling interactions between components that comprise a geospatial product (such as building models, atmosphere, geological formations, intelligent transport networks, and maritime ports) or processes (such as engineering construction processes, geological processes, ocean ecosystems, and navigation) in the field of spatial digital twins (SDTs). This realization naturally leads to the conclusion that some parts of an ideal manifestation of the metaverse will essentially be equivalent to a web-based SDT[16].

These considerations suggest an opportunity to extend the scope of geospatial data and technologies into the metaverse, reaching a wider audience and participating in entirely new and emerging markets. To explore the nature and extent of these opportunities, WGIC conducted a meta-analysis of the existing literature and canvassed a range of SMEs to reveal a consensus of thought-leader opinions.

11 *Delgado et al. 2020, A research agenda for augmented and virtual reality in architecture, engineering and construction, Advanced Engineering Informatics 45, 101-122*
12 *https://www.youtube.com/watch?v=RqEKy3cvRGo*
13 *https://www.vu.city/news/vu-city-virtual-reality-model*
14 *https://connected-environments.org/portfolio/vilo-platform/*
15 *https://kess.org.uk/2018/05/02/prof-shailey-minocha-dr-ana-despina-tudor-ou-role-virtual-reality-geography-science-fieldwork-education/*
16 *https://wgicouncil.org/publication/reports/download-the-wgic-spatial-digital-twins-report/*

4. Definitions of the Metaverse

A new blog or published media article provides yet another view of the definition and application of the metaverse daily. One of the earliest references to the concept in the geospatial domain can be found in the 1997 article[17] by Jonathan Taylor.

Taylor used a metaverse analogy to describe the operation of an SDT of a shopping mall containing virtual stores, virtual goods, and the ability to perform retail transactions using a virtual credit card. The relationship between SDT and the metaverse concept has been re-asserted by prominent geographers[18,19] and WGIC in its report **Spatial Digital Twins: Global Status, Opportunities and the Way Forward**. WGIC has formally adopted the following etymology based on what it believes to be the primary motivation behind the metaverse.

meta:
A prefix meaning beyond, transcending, or more comprehensive, and in this context, like metadata, is self-referential.

universe:
The universe is everything. It includes all of space and all the matter and energy that space contains.

metaverse:
A digital, and therefore, virtual, representation of the universe.

The above definition proposed by WGIC issues a holistic, flexible perspective of the metaverse.

The Open Geospatial Consortium (OGC) and its Geo for Metaverse Domain Working Group (DWG)[20], on the other hand, has adopted the following fairly precise, though still wide-ranging, definition of metaverse proposed by Matthew Ball in his book *The Metaverse: And How It Will Revolutionize Everything*[21].

"The metaverse is a massively scaled and interoperable network of real-time rendered 3D virtual worlds and environments which can be experienced synchronously and persistently by an effectively unlimited number of users with an individual sense of presence, and with continuity of data, such as identity, history, entitlements, objects, communications, and payments.".

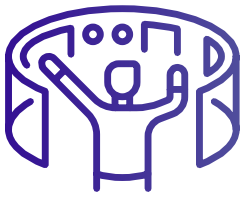
17 *Taylor, J., 1997. The Emerging Geographies of Virtual Worlds, The Geographical Review 87 (2): 172-192*
18 *The Metaverse is Geospatial with Dr Nadine Alameh (Open Geospatial Consortium) https://www.youtube.com/watch?v=VwLY7SxrW14*
19 *Hudson-Smith, A., & Batty, M. (2022). Ubiquitous geographic information in the emergent Metaverse. Transactions in GIS, 26, 1147–1157.*
20 *https://www.ogc.org/press-release/ogc-announces-new-geo-for-metaverse-domain-working-group/*
21 *https://www.matthewball.vc/all/themetaverse*

Some aspects of this highly technical definition resonated with the SMEs who participated in this research. When asked to describe the metaverse and its link with practical uses in the geospatial domain, most of the geospatial SMEs agreed on the following features:

- ✔ It will provide access to a rich array of geospatial data (multi-variate and multidimensional) about the real world in a VR-like user experience,
- ✔ These datasets will span different subject domains and may occur on varied technology (e.g., domain-specific digital twin) platforms but will be interoperable and accessible seamlessly via different modes (e.g., head-mounted displays, smartphones, 2D monitors),
- ✔ VR environments will be enabled with generative type AI and ML to take away the effort to create new entities and aid workflows in the VR worlds,
- ✔ A few SMEs opined that realistic datasets will be updated in real time - requiring a two-eyed, stereoscopic perception of the digital content, providing 'true depth' information, i.e., intuitive, visual distance perception and spatial relationships. However, photorealism for datasets was not seen as a must-have by most SMEs for immersive experiences.
- ✔ The user experience will be highly interactive, and the information flow will be bi-directional, i.e., users will send information into the system, and the system will be able to respond or send information back to the users.
- ✔ Most SMEs reiterated the importance of simultaneous user participation as acentral difference between a metaverse and a standard digital workplace environment.

The disparities worldwide in the state of the enabling technological infrastructure and the state of the economy were also apparent. It is interesting to note that such disparities in the use and access to VR technology were predicted as far back as 1994 by Batty and Barr[22].

- ✔ A few SMEs in emerging economies expressed that the ability for high-quality video conferencing as reasonably sufficient for a definition of the metaverse,
- ✔ For some SMEs, the actual value of the metaverse and its scope were couched in extending the reach of geospatial technology to remote areas for purposes such as education and health.



Disparities exist worldwide in the state of the enabling technological infrastructure and the state of the economy. It is interesting to note that such disparities in the use and access to VR technology were predicted as far back as 1994 by Batty and Barr.

22 *Batty, M., and B. Barr. 1994. The Electronic Frontier: Exploring and Mapping Cyberspace. Futures 26 (7): 699-712.*

5. Building Geospatial Capabilities for the Metaverse

The concepts and technology to create interactive, immersive 3D visualization and datasets suitable for the metaverse have been well-established in the geospatial community for some time.

This section adds further weight to our view that the metaverse will require geospatial approaches, technology, and experience at its foundations. However, it is essential to note that some milestones and challenges already exist or would arise during the construction of the geospatial foundations of the metaverse.

5.1 Strategic Capabilities to Develop the Geospatial Ecosystem in the Metaverse

Several synergies exist between the geospatial domain and the metaverse, where the geospatial offerings naturally address the core needs of the metaverse formation. Here is an enumeration of the existing capabilities of the geospatial domain and the additional ground it needs to cover to build a compelling metaverse. This section should be read together with **Appendix 3**, which lists several products currently in the geospatial industry that serve as the building blocks of the creator economy and spatial computing layers in the metaverse value chain.

a. Online Presence Across Multiple Devices

- ✔ **Metaverse Requirement:** The metaverse will be online and available on multiple platforms for users. It is now widely accepted that transitioning social and work interactions such as teaching, meetings, and retail from in-person to online and multiple platforms is possible and increasingly inevitable. Hybrid working is the new normal. Many critical services, such as health, banking, education, retail, judicial, and governance (at all levels of national/international administration), will continue to move online.
- ✔ **Geospatial Domain Offer:** Due to their size and need for high performance, geospatial data and visualization technologies have traditionally been designed for reasonably powerful desktops (albeit increasingly hosted virtually on a cloud infrastructure) and increased network bandwidth platforms. However, the expertise in mobile geospatial technology (e.g., remote sensing, satellite-based field positioning, handheld field data capture devices/software) related to field operations will provide an excellent starting point for building geo-capable solutions in the metaverse.

b. Data and Technology for Real-time Multi-Player Productivity Scenarios:

- ✔ **Metaverse Requirement:** All interactions and data-sharing between multiple individuals in real time are essential aspects of the metaverse user experience.
- ✔ **Geospatial Domain Offer:** Production scenarios that involve multiple editors/designers in the same or multiple organizations engaged in a singular activity (e.g., Building Information Modelling or BIM) of a new hospital or disconnecting editing and versioning) are relatively well-established in the geospatial industry. A separate chat-like application is used for real-time communication only between editors/and designers. For instance, **Appendix 3** details Oracle and Trimble solutions to enable communication across teams. In the metaverse,

geospatial applications would have to extend this capability to those available in massively multi-player online games (MMOG) to create an authentic replica of the experience of working together in collaboration in the real world.

c. Real-time Interoperability on a Massive Scale

- ✔ **Metaverse Requirement:** Real-world physical and non-physical systems (such as infrastructure, environment, atmosphere) in an ideal metaverse are supposed to be seamless at all geographical scales, thereby allowing a user to create a holistic view of the real world, from street level to global level, through a unified mechanism of command and control.
- ✔ **Geospatial Domain Offer:** Several successful examples of real-time interoperability and required technology in interlinked geospatial systems exist in areas such as SDT, BIM, and integrated transport systems[23]. When highly detailed, the geographical scales of these implementations are usually at the city or an individual entity (e.g., a building or bridge) level. The SDT implementations are generalized on smaller scales for larger systems (e.g., ocean, state, or country). Creating a constantly-evolving, geospatialy-accurate virtual version of interlinked multi-scale real-world physical and non-physical systems is not viable with the existing geospatial technology. The data storage and complicated interactions would overwhelm the infrastructure. The geospatial domain resolves information overload issues by developing suitable generalizations that fit the intended purpose, e.g., London Underground Tube Map[24], BIM levels of detail[25], and sophisticated data compression technologies such as Enhanced Compression Wavelet (ECW), all of which could inspire equivalents in the metaverse.

d. User Control of Content and Creation

- ✔ **Metaverse Requirement:** Web 3.0, i.e., increased user control on the creation and ownership of data, is one of the metaverse's core and attractive features. This aspect is vital to allow trust, openness, collaboration, and creativity in the metaverse. However, enabling these goals will require addressing data quality, privacy, and ethical challenges.
- ✔ **Geospatial Domain Offer:** Concepts such as neogeography (volunteered geographic information or VGI)[26] and quantitative geography have addressed aspects of these subjects extensively. For example, the popular online mapping service OpenStreetMap[27] was originally entirely produced by the contribution of GPS tracks by the public. Most policies based on census geography are based on a suitable aggregation of detailed private information from households.

e. Provision of Automation and Performance using AI/ML

- ✔ **Metaverse Requirement:** The application of AI technologies to support the scale and complexity of interactions in the metaverse will include enabling the creation and maintenance of new user experiences, predicting and monitoring user behavior, data processing, and fully autonomous operation of complex industrial digital twins[28]. For instance, based on generative AI, opus.ai[29] has released demos that show the

23 *https://en.wikipedia.org/wiki/Intelligent_transportation_system*
24 *https://en.wikipedia.org/wiki/Tube_map*
25 *https://www.designingbuildings.co.uk/wiki/Level_of_detail_for_BIM*
26 *Neogeography, 2009. Special Issue Journal of Location Based Services, https://www.tandfonline.com/toc/tlbs20/3/2*
27 *https://www.openstreetmap.org/*
28 *https://www.siemens.com/global/en/company/insights/the-emergent-industrial-Metaverse.html*
29 *https://opus.ai/*

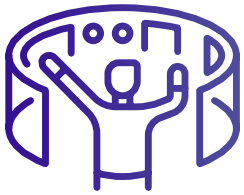
creation of 3D worlds using plain text. NVIDIA has released a no-code tool, Instant NeRF,[30] that can create 3D assets using 2D photos and videos.

- ✔ **Geospatial Domain Offer:** AI/ML technologies are used widely in the geospatial domain for various decision-making applications, e.g., pattern recognition in images, transport network analysis, and vehicle navigation[31].

f. 3D Content and Modelling for Virtual Worlds:

- ✔ **Metaverse Requirement:** In the age of the metaverse, people will be able to see virtual reality in real-world surroundings (e.g., AR/XR/MR using HMD on a construction site), and they will be able to see the real world in a virtual world application (e.g., a detailed 3D model of real infrastructure fully accessible in VR). For the metaverse to be a believable proxy of the natural world, 3D modeling of all structural and functional aspects of real-world entities in natural and built environments will be required.
- ✔ **Geospatial Domain Offer:** The geospatial domain has a long-established ecosystem in reality capturing for use in VR modeling and associated visualization software and hardware. This is evident in the widely available implementation of SDTs, e.g., the City of Victoria (Australia) [32], and those in the making, e.g., XVRS of the NEOM (Kingdom of Saudi Arabia)[33]. The National Underground Asset Register[34] project demonstrates how the information on a complicated network of underground assets (e.g., utilities) spread across an entire city can be accurately recorded as an SDT. Refer to **Section 4** for more examples of geospatial software and solutions.

Table 1 provides a strategic outlook based on a PESTLE-SWOT matrix for building geospatial capabilities of the metaverse concept. The matrix lists the internal strengths and weaknesses of the geospatial ecosystem, the risks that must be explored and the opportunities the stakeholders must navigate.



The metaverse has the potential to create new economic opportunities. Some creators and entrepreneurs within virtual worlds were generating substantial income, with top earners making millions of dollars in virtual transactions and sales.

30 *https://blogs.nvidia.com/blog/2022/03/25/instant-nerf-research-3d-ai/*
31 *WGIC, 2021. Geospatial AI/ML (GeoAI) Applications and Policies – A Global Perspective, https://wgicouncil.org/publication/reports/wgic-geoai-report-geospatial-ai-ml-applications-policies-global/*
32 *Digital Twin of Victoria, Australia, https://www.land.vic.gov.au/maps-and-spatial/digital-twin-victoria*
33 *https://cognitive.neom.com/insight/neom-tech-and-digital-co-unveils-xvrs.html*
34 *https://www.gov.uk/guidance/national-underground-asset-register-nuar*

Table 1: A SWOT-PESTLE strategic analysis[35 36 37]

	Strength	Weakness	Opportunities	Threats
Political	• There is increasing awareness of and support for the role of geospatial technology and ecosystem in the metaverse within the political establishment.	• Some governments are primarily interested in financial trading and gaming-type applications in the metaverse.	• A great deal of support exists from governments in Asia and the Middle East. • The growing awareness of this market within the rest of the significant economies provides an opportunity to set up stakeholder engagement early.	• A few governments want to regulate development heavily due to concerns about cyber security, personal safety, national interest, and privacy.
Economic	• A metaverse is essentially a web-based geospatial ecosystem. Geospatial ecosystems have well-established business models to drive ROI and value for money.	• Given the long lead time for the maturity of the metaverse and the ROI of concepts like digital twins, geospatial SMEs find it challenging to raise funds for R&D.	• Metaverse will be a global phenomenon. Based on information about how much it costs to make a small SDT, the market size predicted by industry analysts for a platform of SDTs is likely vastly underestimated.	• Some thought leaders have an intense desire to keep the metaverse open and accessible to all. This could potentially require elaborate revenue-sharing approaches.
Social	• Technologies such as TomTom, SatNav and Bing Maps have made geospatial data and interfaces a daily routine for traveling and information gathering.	• Using sophisticated AI to augment some geospatial capabilities would lead to societal tensions.	• Just as WWW did, the metaverse will democratize access to geospatial knowledge and services to the broader public.	• Market tension exists for seeking a prominent position in the metaverse narrative, and geospatial lacks broader public appeal, e.g., compared to the gaming industry. • It remains unclear how a balance between the need for personal privacy and system requirements of relevant data to represent reality would be achieved.
Technological	• AR/MR/VR/XR are well established in the geospatial domain. • SDT and BIM have proven geospatial ecosystem capability to construct functional equivalents of complex real-world applications.	• Although far less now, there has been an underappreciation of the complexities of the geospatial ecosystem in the mainstream IT industry.	• Engagement with like-minded stakeholders in computer graphics, social media, and computing infrastructure communities will provide R&D opportunities for novel solutions to technological challenges in building the metaverse.	• Management and governance of data, technology, Intellectual property rights, and accessibility will require extensive thinking to achieve a desirable level of openness and inclusivity.
Legal	• Most geospatial bodies navigate these topics routinely	• The legal landscape to regulate geospatial data licenses across domains can be complicated.	• Participation in forums such as the World Economic Forum, UNGGIM, and Metaverse Forum would provide access to large groups of stakeholders at the highest level to aid the navigation of legal complexities.	• A few governments want to regulate heavily.
Environmental	• Long-established track record of using geospatial ecosystems in decision-making, analyzing, and communicating environmental issues.		• Several studies have proposed the value of the metaverse and the associated geospatial technologies, such as SDTs, for fulfilling the UN Sustainable Development Goals.,	

35 *https://initiatives.weforum.org/defining-and-building-the-metaverse/home*
36 *https://sdgs.un.org/sites/default/files/2022-05/2.1.4-27-Umar-Metaverse4SDG.pdf*
37 *https://www.youtube.com/watch?v=enx4lAkB9GQ*

5.2 Next Steps for Embedding Geospatial Capabilities into the Metaverse

A multifaceted roadmap is required to fully integrate the geospatial capabilities into the metaverse and benefit from the possibilities of emerging web structures. Some milestones require engaging with topics that are generally not widely discussed within the geospatial industry. These issues include political drivers, social concerns of the public and governments, value propositions to investors and companies, and technological challenges such as hardware, data, software, and governance interoperability. A practically helpful parallel version of the real-world geographical context for metaverse users can only be achieved through some shared global objectives on these themes.

The list of priorities enumerated below for the proposed next steps does not necessarily have any sequential order and, therefore, could be developed in parallel. The intended recipients of priorities range from the organizational leaders responsible for senior-level stakeholder engagements and market development to technical managers and developers.

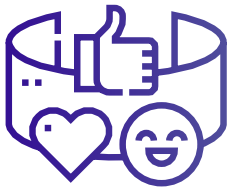
Priority 1 – Understand and steer societal perception

Engagement with the general public

The metaverse promises to fundamentally change how we live and work. Managing the perception of this change to promote the positive value proposition of the metaverse will require a careful understanding of the participant's perception of the change and their aspirations.

Nearly two-thirds of respondents amongst the general public in the US and the UK, especially the younger generation (18-44 years old), reported being aware of the metaverse[38] in a 2022 YouGov poll[39]. At the same time, nearly two-thirds were not confident about explaining the metaverse. Less than 20% of respondents would have chosen the metaverse as a work environment and indicated a preference to use the metaverse for gaming, socializing, and entertainment.

Previous mass marketing campaigns have created a general perception that the metaverse will primarily be immersive and accessible via high-end computing platforms, VR, and HMDs. Opinions from the YouGov survey and some industry leaders[40] reveal apprehensions about adopting such a vision because it is considered anti-social



The metaverse's transformative potential necessitates understanding and shaping public perception. A 2022 YouGov poll found younger generation in US and UK being aware of metaverse. But less than 20% of respondents wanted to choose metaverse as a work environment and indicated a preference to use it for gaming, socializing, and entertainment.

38 *For this study, YouGov defined the Metaverse as a virtual world where you can create an electronic version of yourself and then take part in activities on your devices and associated equipment.*

39 *https://business.yougov.com/sectors/technology/2022-uk-us-metaverse-report*

40 *2023, AR vs. VR: Dueling Realities in the Metaverse Arena with Nokia's Head of Innovation Leslie Shannon, https://www.youtube.com/watch?v=A2EMHPf--BY*

and could create barriers to personal communication. An example is the failure of Google Glass[41], aimed at the mass market, to gain a market stronghold.

This belief likely stems from a combination of a poor impression of the current state of technology and personal preference. However, public perception of HMDs being anti-social could evolve across different geographical regions and necessities. For example, wearing face masks has long been considered routine in public places in Japan. However, until the COVID-19 pandemic, wearing face masks in much of the world would have created awkward social situations and raised suspicion when passing through a security check. But during and after the pandemic, face masks have become much more widely accepted.

The uptake of HMDs would vary across geographical regions due to local cultural norms and affordability. The public perception of the metaverse and associated technology in Asian countries is very different, as would be evident from substantial investment and government support in China, Japan, Singapore, and South Korea. Therefore, HMDs will appeal to all users in Asia; however, they will likely remain more successful in work and gaming use cases in the West.

Government Policy, Strategy, Legislation and Regulations

Data privacy, intellectual property rights, illegal behavior, transparency of commercial transactions, market competition, and preservation of national cultural identities on web platforms are already a concern for most public and commercial organizations. The increasing use of and improvement in AI (e.g., deep fakes, generative text) on the metaverse web platforms will make it even harder to distinguish between the activities of a well-intentioned person, a fraudster, and a bot. Therefore, several national governments and expert bodies are developing regulations and policies to govern these next-generation web-based worlds. As is happening now, some of these regulations may create additional administrative overhead for companies whose business model relies on more liberal or fewer regulations.

Therefore, it will be imperative for the entrants to engage with relevant government bodies and affiliated stakeholders in these regions to understand the growth areas and develop ways of working with them.

In 2022, China's Ministry of Industry and Information Technology (MIIT) announced a four-year action plan[42] for integrating and developing VR and industry applications. The focus of this plan included tactical activities such as creating suitable business environments for creator and manufacturer ecosystems, expanding new applications,



Public perception and acceptance of technologies such as head-mounted displays (HMDs) vary across regions due to cultural norms and affordability. Asian countries, such as China, Japan, Singapore, and South Korea, have displayed a more favorable view of the metaverse, leading to substantial investment and government support.

41 *https://en.wikipedia.org/wiki/Google_Glass*
42 *https://www.gov.cn/zhengce/2022-11/01/content_5723274.htm*

interoperability/standardization, and strategic activities such as setting up plans to maintain a steady talent supply. Examples of activities leading on from this action include the metaverse[43] policies by Guangzhou Huangpu District and Guangzhou Development Zone and the launch of a metaverse technology industrial patent pool[44] set up by Beijing Intellectual Property Operation and Management Co., Ltd. and Baidu.

In 2023, the European Commission (EC) announced its strategy[45] on Web 4.0 (an extension of Web 3.0 in which physical and digital worlds seamlessly blend, enabling intuitive and immersive experiences) and the next generation of virtual worlds. In addition to creating a creator economy and conducive business environment, a vital objective of the strategy is to limit the potential for organizations to dominate by ensuring that global standards are open and inclusive.

In 2022, Japan's Ministry of Economy, Trade and Industry (METI) set up a "Web 3.0 Policy Office" to bring together the departments responsible for industrial finance, taxation, corporate system (vehicles), business environment issues, media and content, sports, fashion, and the other related industries. METI considers the metaverse as the new personal interface in digital spaces for the younger generation. It recognizes the need for governing the enormous business potential and associated risks. As part of this initiative, a research project has been set up[46] to identify legal and other business environment barriers to entering the creator economy market for Japanese creators, support creator economies, and set up stakeholder engagement.

South Korea's Ministry for Science and Technology (MSIT) announced a three-year plan in 2022 to create the Metaverse Industry Promotion Act to govern MSIT's interest and bolster MSIT's investment in the metaverse. MSIT published eight ethical principles for the metaverse[47] : authenticity, autonomy, reciprocity, respect for privacy, fairness, personal information protection, inclusiveness, and responsibility for the future. A key rationale behind the laws and principles is their belief that gaming-related rules are not directly translatable to the metaverse.

In 2023, the UK Government announced[48] that its Department of Science, Innovation and Technology (DSIT) would lead the future of the metaverse in the UK. Much like the MSIT in South Korea, the DSIT strategy focuses on developing potential economic growth opportunities for the creator economies and studying the regulations



Recognizing the metaverse's importance, Japan's Ministry of Economy, Trade and Industry (METI) set up "Web 3.0 Policy Office" in 2022 to bring all concerned departments together, remove barriers for Japanese creators, support creator economy and engage with stakeholders.

43 *http://www.hp.gov.cn/gzjg/qzfgwhgzbm/qgyhxxhj/xxgk/content/post_8171935.html*
44 *http://bj.people.com.cn/n2/2023/0531/c14540-40439083.html*
45 *Generation Virtual Worlds: Societal, Technological, Economic and Policy Challenges for the EU, Publications Office of the European Union, Luxembourg, 2023, doi:10.2760/51579, JRC133757, https://publications.jrc.ec.europa.eu/repository/handle/JRC133757*
46 *https://www.meti.go.jp/english/press/2022/0823_002.html*
47 *https://www.msit.go.kr/bbs/view.do?sCode=eng&mId=10&mPid=9&bbsSeqNo=46&nttSeqNo=17*
48 *https://www.gov.uk/government/publications/spring-budget-2023/spring-budget-2023-html*

on data privacy and online safety. An all-party parliamentary group[49] is expected to research a broad spectrum of topics (investment, taxation, education, diversity, safety, and others) under the umbrella themes of Web 3.0 and metaverse. The group is expected to provide evidence, guidance and recommendations to MPs and Lords in the Parliament. Although the UK Geospatial Strategy 2030[50] doesn't directly reference Web 3.0 and metaverse, its mission includes building the country's capabilities on location data and AI and shaping their use in infrastructure, security, and land use applications.

In 2022, the United Arab Emirates (UAE) announced the Dubai Metaverse Strategy[51] to create 40,000 virtual jobs by 2030. The strategy primarily focuses on building the technology and infrastructure for blockchain and metaverse economies. As part of this initiative, the Dubai International Financial Centre (DIFC) has set up DIFC Metaverse Platform, which includes three main initiatives – a) An accelerator program with a dedicated physical studio for metaverse technology, which will look to promote the development of a creator community and venture building; b) policy development in the metaverse sector, legislation on open data, digital identity and company law frameworks in the metaverse; and c) development of a metaverse community that will explore ways to enhance the metaverse experience for customers. Dubai's Virtual Asset Regulatory Authority has announced its presence in the Sandbox Metaverse platform to test the use of the regulatory framework and use cases of metaverse payment mechanisms.

Other significant economies such as The United States[52], India, Singapore, and Australia[53] allude to growing attention to metaverse through social media and government announcements. Still, there is no formal policy and strategy at the time of writing this report.

These policy-related engagements have to focus on, among others, non-technical interoperability, e.g., jurisdictional interoperability or regulatory interoperability (World Economic Forum)[54]. These discussions must consider the stakeholders' varied locations, regulations, and regional and cultural expectations.

Priority 2 – Setup Stakeholder Engagement and Collaboration

Given its multifaceted and global nature, understanding the wide variety of stakeholders involved in the metaverse value chain and end user communities would be vital in identifying the avenues of collaboration the geospatial community can pursue. Using the Boston Matrix approach for stakeholder mapping, we can identify four stakeholder groups for the metaverse:



In 2022, the United Arab Emirates (UAE) introduced the Dubai Metaverse Strategy, aiming to generate 40,000 virtual jobs by 2030. The strategy includes an accelerator program, policy development for the metaverse sector, and the creation of a metaverse community.

49 https://appgmetaverseweb3.com/
50 https://www.gov.uk/government/publications/uk-geospatial-strategy-2030/uk-geospatial-strategy-2030
51 https://u.ae/en/about-the-uae/strategies-initiatives-and-awards/strategies-plans-and-visions/government-services-and-digital-transformation/dubai-metaverse-strategy
52 https://crsreports.congress.gov/product/pdf/R/R47224
53 https://www.standards.org.au/documents/h2-3061-metaverse-report
54 https://www3.weforum.org/docs/WEF_Interoperability_in_the_Metaverse.pdf

Type 1 Influence and Keep Satisfied: These stakeholders are likely to have great importance/authority but may only have nominal interest in the metaverse due to several competing priorities. These stakeholders include governmental and non-governmental organizations that create regulations, policies, and guidelines. Some examples of stakeholders include the US Congress, European Commission, MSIT (South Korea), DSIT (UK), MIIT (China), and METI (Japan).

Type 2 Collaborate Closely: These stakeholders are essential, influential, like-minded, and interested in developing and implementing the metaverse value chain. These stakeholders are involved in the research, development, and marketing of the metaverse, with a focus on software, content, platforms, technologies, and standards. This group will host the skilled resources and knowledge required to create and maintain the metaverse. Examples include the Metaverse Standards Forum, Geo for Metaverse DWG, Khronos Group[55], International Standards Organisation (ISO), British Standards Institute, The Virtual and Augmented Reality Industrial Coalition[56], and Digital Twin Hub[57].

Type 3 Consult: These stakeholders are keen supporters of the metaverse but may not themselves be involved in the actual implementation. These stakeholders can be ideal listening posts and promotional platforms to seek out information to gather opinions and requirements. Examples of these stakeholders include dedicated industry groups and forums on metaverse and digital twins.

Type 4 Keep Informed: This group of stakeholders are currently not interested in metaverse technology due to various constraints. In its mature form, the metaverse will likely arrive at quite different points worldwide. For example, stakeholders in emerging economies and areas away from technology hotspots will have to be kept informed to create opportunities for future market growth and to adapt the metaverse solution to the specific technological capabilities of these communities. Activities with these stakeholders provide information to develop an understanding of "usage interoperability"[58], which is the ability for different demographic groups to participate in the metaverse across different geographies at low cost, leading to inclusion without any discrimination.

The experts who participated in this study recommended engagements with Type 1 and 2 stakeholders as the immediate next steps for improving stakeholder engagement.

Priority 3 – Investigate complexities of technological interoperability and data governance

Technological complexities such as the interoperability of SDTs in bringing a geographical context to the metaverse are perhaps the most well-understood of all future issues. They are currently being addressed by several industry consortiums and individual solutions providers. Two prominent examples of such forums, where active participation by the geospatial community would be vital to stay up-to-date and provide inputs, include:

- ✔ The Metaverse Standards Forum (MSF), comprised of over 2000 organizations from various IT domains, including geospatial, aims to coordinate cooperation amongst a constellation of international standards organizations. The geospatial domain is yet to be assigned the status of a fully working group[59] at the time of writing this report and needs to be taken up by the community and other stakeholders.

55 *https://www.khronos.org/*
56 *https://digital-strategy.ec.europa.eu/en/policies/virtual-and-augmented-reality-coalition*
57 *https://digitaltwinhub.co.uk/*
58 *https://www3.weforum.org/docs/WEF_Interoperability_in_the_Metaverse.pdf*
59 *https://metaverse-standards.org/news/blog/metaverse-standards-forum-announces-first-approved-domain-groups/*

✔ Together with the MSF, the Geo for Metaverse DWG at the OGC has focused on developing interoperable data formats such as Universal Scene Description (USD), sometimes called HTML of the metaverse.

These early initiatives set up the foundation of workflows needed to evolve the current SDT and BIM systems into competent and practically helpful components of the metaverse[60] (**Figure 5**).

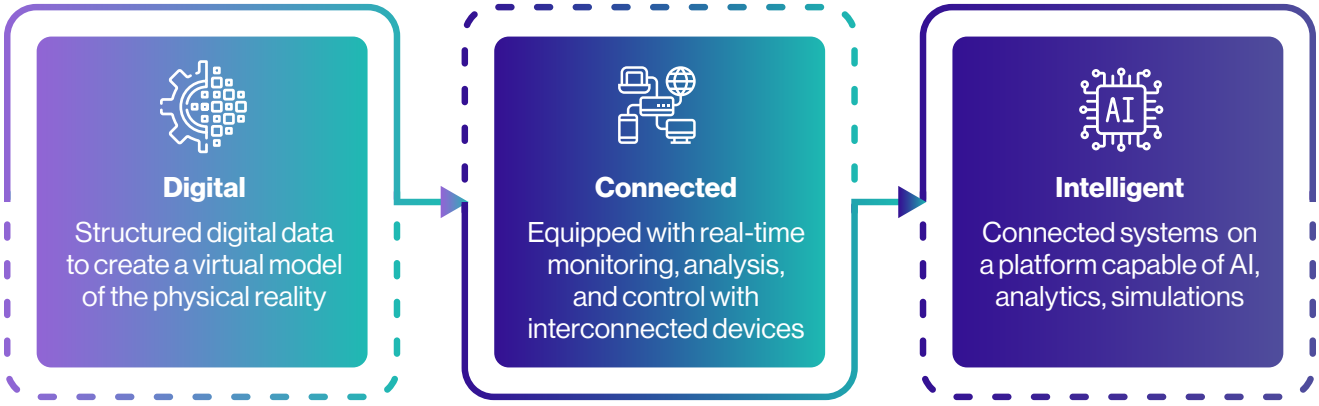


Figure 5: Expected evolution of geospatial systems to become part of metaverse
(Source: Adapted from Hexagon view on urban digital twin)

Data Interoperability across the Metaverse

There are drives to improve the interoperability of several aspects of data (e.g., ontology, schema, quality) within specific domains (e.g., urban, marine) in various industrial (e.g., Overture Map Foundation Data Schema[61]; see **Appendix 4**) and research communities. There are research areas that remain relatively unknown and should be explored to identify gaps in knowledge that need to be addressed. These would require the community to tackle interoperability at a conceptual level across domains instead of particular data formats and exchange protocols.

✔ **Universal Digital Twin (UDT)**[62] is a dynamic knowledge graph approach that uses semantic web technology. It allows the creation of distributed cross-domain interoperability and ensures that data are connected, portable, discoverable, and queryable via a uniform interface. A practical example of a UDT is the UK Digital Twin,[63] which demonstrates how distributed knowledge graphs and collaborative, intelligent agents share, analyze, and combine data to answer complex questions related to land use, gas transmission, electrical power, and flooding. As a next step, other types of SDTs (e.g., weather, transport, health) could be integrated to test the scalability of the UDT concept.

✔ There are several scenarios of dataset integrations/linking in which, despite similarities between candidate data fields to be joined, the actual information represented by the joined data fields from the datasets may only be partially related. For example, the geospatial extent associated with a postal address database can be quite different from the geospatial extent related to the land ownership database, or the topographic scales used for words hill, forest, or city could have widely different ranges in different parts of the world. A limited understanding

60 *https://hexagon.com/go/sig/urban-digital-twin*
61 *https://docs.overturemaps.org/*
62 *Akroyd, J., Mosbach, S., Bhave, A., & Kraft, M. (2021). Universal Digital Twin - A Dynamic Knowledge Graph. Data-Centric Engineering, 2, E14. doi:10.1017/dce.2021.10*
63 *https://kg.cmclinnovations.com/explore/digital-twin*

of such a nuanced relationship can lead to misleading outcomes. The UK Geospatial Commission's Correlation Relationship[64] project demonstrated how an agreed ontology and a data model could be used to store and distribute the partial relationships between a small number of datasets. As a next step, other datasets from different digital twins should be explored for such relationship scenarios.

Quality of Visualization

- ✔ The quality of graphics reported by the users of Meta's Horizon Worlds[65] is considered one of the reasons behind the initial setback in the launch. Interestingly, the experts in this study did not consider photorealism as a must-have for creating immersive experiences and deep engagement. However, the experts indicated the need for an acceptable quality rendering of geospatial data, particularly for gaming or education-like applications (e.g., virtual geology field trips), in which the level of detail could be an essential aspect of learning content. Therefore, extensive user research is needed to establish the acceptable quality of geospatial rendering in various application contexts and display form factors (i.e., smartphones, HMDs, desktops, and large displays).
- ✔ Unless the social stigma and problematic ergonomic factors of current HMD designs are mitigated, their adoption would largely be limited to short duration in gaming, industrial metaverse, and AEC-type workplaces for AR/MR/XR use. This limitation of HMDs could also be a much-needed catalyst for novel approaches for new designs of *metaverse-capable* built structures equipped with discretely located IoT/NFC sensors, holographic interfaces, and projection systems for full immersion, and form factors of HMDs reduced to those of smart contact lenses developed by Google[66], Mojo[67], and Samsung[68] under development. Therefore, given that HMDs are widely used in geospatial communities, Proof of Concept (PoC) could be carried out with like-minded partners in the computer graphics, architecture, and product design communities to work on these barriers.
- ✔ The vision of the metaverse is to be omnipresent, i.e., to allow users to dip in and out of the metaverse in any walk of life on any device form factor. A commonly cited example of this capability is how our phones can switch automatically from indoor Wi-Fi to external cellular networks. Geospatial data and technology have traditionally been couched in comparatively robust computing environments, whether powered through a local CPU or a virtual one in the cloud. It is generally believed that early versions of the metaverse will mainly be accessed using smartphones/tablets in outdoor spaces or by a conventional desktop monitor setup indoors. While it is possible to undertake AR/VR/MR/XR activities outdoors using small form factors, the computing platforms and supporting telecommunication infrastructure are still far from allowing a seamless global transition for complex geospatial ecosystems. Therefore, user research is needed to develop UI, data simplification, and rendering approaches, particularly on smartphone platforms, almost in preference to conventional desktop form factor.

Geospatial Data Governance

Given that a well-established geospatial data ecosystem exists for producing high-quality multidimensional geospatial datasets in various formats and platforms, SMEs consulted in this study did not consider the production

64 *https://geospatialcommission.blog.gov.uk/2020/04/23/building-correlations-between-geospatial-datasets/*
65 *https://www.meta.com/en-gb/experiences/2532035600194083/*
66 *https://www.youtube.com/watch?v=k_LfohzPZ1g*
67 *https://www.youtube.com/watch?v=d2fBBJRjccs*
68 *https://medium.com/chip-monks/samsung-patents-smart-contact-lenses-with-built-in-cameras-dc3a1bd0600f*

of geospatial datasets as a data-related milestone. However, experts highlighted the need to address the governance issues listed below by creating seamless access between real-world and real-time systems on the metaverse.

Most of the issues belong to a type in Q-FAIR,[69] i.e., Quality, Findability, Accessibility, Interoperability, and Reusability. These issues have not arisen solely due to the metaverse but have achieved a new, much higher level of exposure due to the much higher level of access.

- ✔ Given the strong desire to create an open, free metaverse with the most comprehensive outreach, business concerns on intellectual property rights (IPR) on the geospatial data must be addressed.
- ✔ At present, most high-quality national geospatial datasets are procured and shared with specific licensing conditions. Some sensitive geospatial datasets are not shared at all. Data-sharing agreements must be formed between organizations (locally, nationally, and internationally) with their respective legal frameworks to create integrations.
- ✔ Geospatial datasets in each organization have different criteria for acceptability, often expressed using accuracy and completeness targets. The quality of datasets influences the confidence in processes that use the datasets, and some of the processes would have legal consequences. For example, an inaccuracy of a real-world property boundary in the VR world in the metaverse may lead to legal complications and reputation risk. In this case, the geospatial community's experience from AR/MR/XR can be used to develop best practices.
- ✔ The creation of data dictionaries, shared ontologies, metadata standards, and interoperability layers will be required to support integrations.
- ✔ If the metaverse is indeed to be fully open and inclusive, complex appropriate role-based access controls and retention policies will be needed to protect personally identifiable datasets (e.g., travel histories) for personal safety but also in work scenarios.

Priority 4 - Create economic opportunities and market readiness markers

It is not straightforward to extract the market size of the geospatial ecosystem bundled within the predictions on the metaverse market size from industry analysts and the commitments from investments from some of the early adopter national governments. However, it will likely be a few orders of magnitude over the current SDT market versions[70]. There are also open business model-related questions on sharing the cost to create, maintain, and govern the metaverse. Several milestones, as suggested below by the experts, therefore, would have to be met:

- ✔ To kick-start momentum, a unique value proposition, even if simple, must demonstrate the value of bringing geospatial context to the metaverse. This is crucial for creating a critical mass of initial investors and owners of the metaverse infrastructure to achieve a build-up to widespread adoption.
- ✔ Approaches on symbiotics in business models (e.g., public-private partnership) and other revenue generation in aspects of geospatial ecosystems (e.g., data licensing, IPR) have to be developed for a tangible and enduring

69 *https://geospatialcommission.blog.gov.uk/2021/06/25/byte-ing-back-better-introducing-a-q-fair-approach-to-geospatial-data-improvement/*

70 *According to Mordor Intelligence, the digital twin market was valued at $10.27 billion in 2021, and it is expected to reach a value of $61.45 billion by 2027, registering a CAGR of 34.48% over the forecast period, 2022-2027. Source: https://geospatial.trimble.com/en/resources/blog/demystifying-spatial-digital-twins*

return on investment (ROI). For example, Digital Twin of Victoria (covering an area of over 227,000 sq. km) in Australia is spending $24 million, Virtual Singapore (covering an area of 729 sq. km) Digital Twin cost $53 million, and Dubai Digital Twin (covering an area of 4110 sq.km) has cost $20 billion[71]. These cost variations indicate differences in the capabilities and capacities of these digital twins and thus influence the period for fulfilling the ROI estimates.

- ✔ Citing the current stage of the metaverse market, several SMEs surveyed in this study indicated risk aversion to investing their funds and time in developing ideas and offerings. They recommended the development of a metaverse-readiness indicator that SMEs can use to know at a glance the maturity of global markets that are ready for innovation and investment. This would be similar to the Ball Metaverse Index[72] that ranks and tracks the performance of metaverse companies.
- ✔ Concerns about the adverse impact on economic productivity caused by the use of a gaming-like environment in workplaces would have to be carefully studied to avoid reluctance by employers.



71 https://storymaps.arcgis.com/stories/4ae54ecb2d8e4640b491de1fc319cffc
72 https://www.ballmetaverse.co/methodology

6. Appendices

Appendix 1

Examples of Early Implementation of the Metaverse

Here are some highly regarded and popular metaverse-like platforms. These do not usually have any dedicated links to the geospatial ecosystem as yet.

Platform	Developer	URL
CryptoVoxels	Noland Consulting Ltd.	https://www.voxels.com/
Decentraland	Decentraland Foundation	https://decentraland.org/
Hadean	Epic Games	https://hadean.com/
Horizon Worlds	Meta	https://www.meta.com/gb/en/horizon-worlds/
Mesh	Microsoft	https://www.microsoft.com/en-us/mesh
Omniverse	NVIDIA	https://www.nvidia.com/en-gb/omniverse/
Roblox (Game)	Roblox Corporation	https://www.roblox.com/
Starline	Google	https://blog.google/technology/research/project-starline/
The Sandbox (Game)	Pixowl, Inc.	https://www.sandbox.game/en/

Appendix 2

Definitions of the Metaverse in Industry and Academia

This section summarizes various thoughts from industry experts and academic researchers around the definition of the metaverse. It offers an approach to position these disparate opinions into three groups. Such a positioning will prove informative for decision-makers to manage the public's and investors' expectations.

Group 1 - Thought Leaders

Tony Parisi, one of the prolific thinkers and pioneers in the metaverse community, proposed seven rules that a true metaverse should cohere with, i.e., not necessarily prescribe to[73].

Rule #1	**There is only one metaverse:** Metaverse is the total of all publicly accessible virtual worlds, real-time 3D content and related media connected on an open global network, controlled by none and accessible to all.
Rule #2	The metaverse is for everyone, as defined by our broad societal rules of inclusion. This is not a political or socioeconomic statement but an ethnographic one with political and socioeconomic implications.
Rule #3	**Nobody controls the metaverse:** It is the universal commons for digital communication and commerce, intermediated as needs dictate, governed as required for the common interest, toward the greatest good for the most significant number.
Rule #4	The metaverse is open and is built upon interoperable technologies and tools, connected via rigorously defined and broadly agreed-upon free and open communication standards.
Rule #5	The metaverse is hardware-independent and accessible on any device regardless of display type and form factor.
Rule #6	The metaverse is a network connecting the world's publicly accessible virtual experiences, real-time 3D content and related media.
Rule #7	The metaverse is the Internet, enhanced and upgraded to consistently deliver 3D content, spatially organized information and experiences, and real-time synchronous communication.

73 *https://medium.com/meta-verses/the-seven-rules-of-the-Metaverse-7d4e06fa864c*

Matthew Ball, author of the book *The Metaverse: And How It Will Revolutionize Everything*, provided an all-rounder and precise definition based on nine characteristics of a *metaverse*:

"The metaverse is a **massively scaled [1]** and **interoperable [2] network [3] of real-time [4] rendered 3D [5] virtual worlds [6]** and environments which can be experienced **synchronously [7]** and **persistently [8]** by an effectively **unlimited number of users [9]** with an **individual sense [9]** of presence, and with **continuity of data [8]**, such as identity, history, entitlements, objects, communications, and payments."

Beamable and The Metaverse Insider (A metaverse technology promoter, advisor, and market intelligence company) have created a taxonomy for the metaverse value chain to identify the leading enabler technologies and most companies that are most active in these technologies. This approach to defining the metaverse provides a higher-level structural composition rather than end user technologies. Most notably, the Seven Layers of the Metaverse Value Chain[74] (**Figure 6**) by Beamable identifies geospatial mapping as an integral aspect of spatial computing[75]. According to Radoff, spatial computing aims to allow a seamless bi-directional exchange of information between a user in the real world and the components of the 3D world. It uses real-time (e.g., from IoT sensors) data on 3D spaces and software (e.g., 3D game engines, machine learning, novel UI). The Market Map (**Figure 7**) by Metaverse Insider, although it does not explicitly mention the presence of geospatial mapping in a metaverse

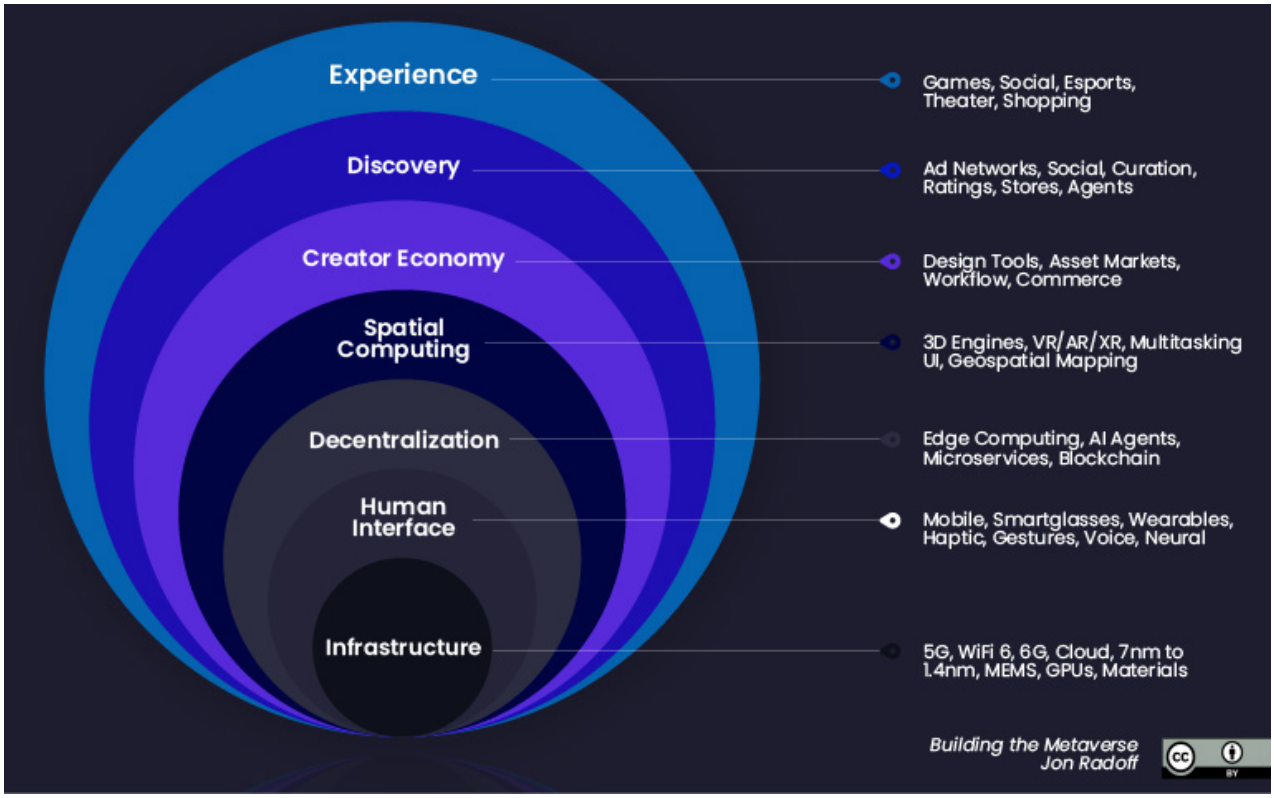


Figure 6: Seven layers of the metaverse value chain (Source: Beamable)

74 *https://medium.com/building-the-Metaverse/the-Metaverse-value-chain-afcf9e09e3a7*

75 *Spatial Computing term has been attributed to Simon Greenwold based on his use of the term for his 1995 MSc Thesis at the MIT; https://acg.media.mit.edu/people/simong/thesis/SpatialComputing.pdf*

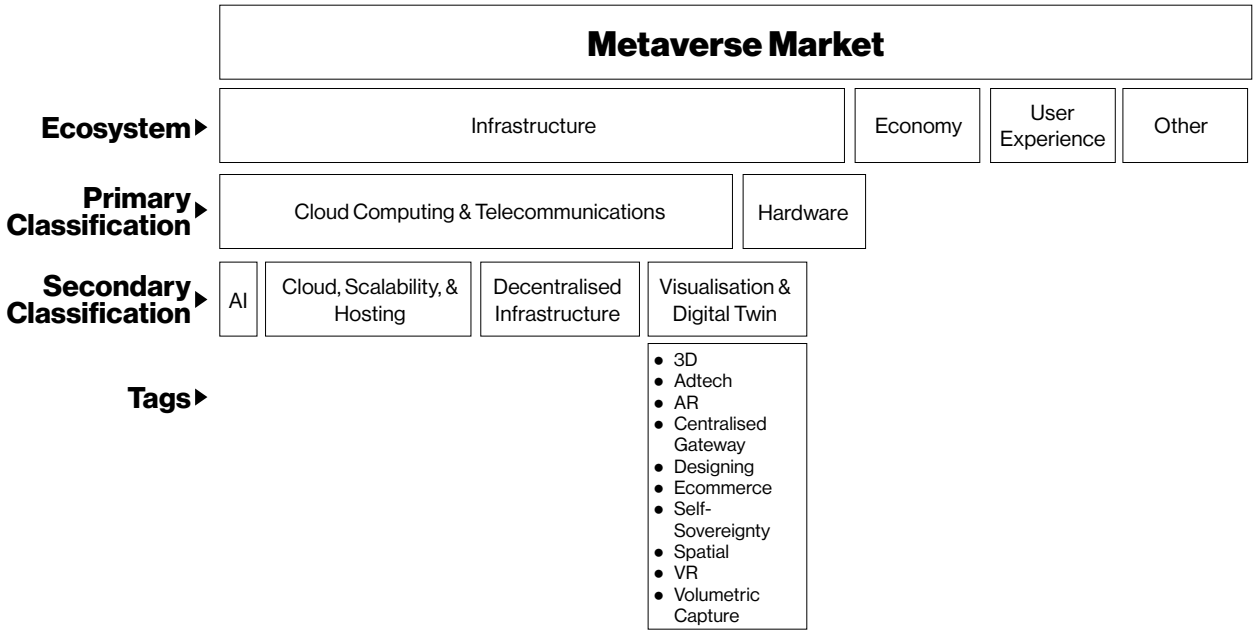


Figure 7: Taxonomy of metaverse market to show the position of geospatial ecosystem components in the market. Note that only infrastructure ecosystem has been elaborated. (Source: The Metaverse Insider)

ecosystem, does identify visualization and digital twin as part of the geospatial components that serve as the real-time digital counterpart of a physical object or process.

Group 2 - Organizations

In the wider IT domain, Neil Trevett, Chair of the prominent Metaverse Standards Forum (MSF)[76] formed in 2022, suggested[77] that "metaverse is an evolution of the web", and it "combines the connectivity of the web with the immersiveness of spatial computing".

In the geospatial ecosystem, OGC has adopted a widely popular definition of the Metaverse by Matthew Ball. At the UN-GGIM, the 2020 future trends in geospatial information management report[78] recognized the growing importance of VR in the digital twin and AEC industry. At the 12th Session of the Committee of Experts in 2022, the term "Geoverse"[79] was proposed for an aspirational globally interconnected geospatial information ecosystem – one that permits intelligent interactions between spatial data infrastructure (SDI) web portals, systems, sensors, applications, devices, and other things using a broad range of communication interfaces and machine facilitated technologies such as AI, ML, Natural Language Processing (NLP), data mining, virtual assistants, digital identities, and blockchain.

The UN-GGIM proposed that the geoverse would share many elements of the metaverse (e.g., 3D, openness, and inclusiveness). However, it would be a superset of the metaverse because geoverse would "extend the notion of a

76 *https://metaverse-standards.org/*
77 *Neil Trevett, 2022, Evolving an Interoperable Metaverse, https://youtu.be/E87y_a14H0I*
78 *UNGGIM, 2020, Future Trends in geospatial information management: the five to ten year vision - Third Edition, https://ggim.un.org/documents/DRAFT_Future_Trends_report_3rd_edition.pdf*
79 *UNGGIM, 2022, Determining the future geospatial information ecosystem, https://ggim.un.org/meetings/GGIM-committee/12th-Session/documents/E-C.20-2022-6-Add_1_Future_Geospatial_Ecosystem.pdf*

3D virtual society to include 4D visualizations, predictive analytics, and real-time knowledge in all its forms, as well as a wide range of integrated and interoperable data from across various sectors and disciplines".

Group 3 - Academic Discourse

The wildfire-like spread of definitions and themes on the metaverse has been a source of much intellectual debate and intrigue to all scientific and non-scientific academic communities. Academic researchers have developed two main approaches for making sense of the rapid influx of information. The first approach proposes that a common structural definition can be found by converging the numerous proposals. The second approach proposes that the definition of metaverse should be seen as ever-evolving and consider competition amongst various stakeholders to position their definition.

Common Denominator Definition

In the first approach to metaverse definitions, there is a belief that a sensible definition can be found by systematically assimilating all known definitions and analyzing commonalities among them. Ritterbusch and Teichmann[80] used a Systematic Literature Review[81] and Affinity Aggregation[82] approach to study the 28 definitions of metaverse in 381 publications dating from 1997 to 2022 from various scientific and non-scientific disciplines. They calculated the frequency of keywords found in these definitions and stated that a combination of these definitions can be expressed as below.

Metaverse, a crossword[sic] of "meta" (meaning transcendency) and "universe", describes a (decentralized) three-dimensional online environment that is persistent and immersive, in which users, represented by avatars, can participate socially and economically with each other in a creative and collaborative manner in virtual spaces decoupled from the real physical world.

Ever Evolving Definition

Instead of attempting to identify a common and stable definition, the second approach to defining the metaverse expects that as technologies change, definitions of the metaverse will continue to evolve based on various, sometimes competing, narratives and that there is, therefore, little value in aiming to reach a consensus definition. Dolata and Schwabe[83] utilized the principles in the Theory of "Social Construction of Technology[84]" (SCOT) to review over 270 news stories (dating 2006-2022). They proposed a detailed matrix of perspectives and social groups defining the metaverse. SCOT theory essentially states that understanding and evolution of technologies go through phases of "sensemaking"; they drive a degree of public speculation where the dominant discussion is how to attempt to make sense of the phenomenon. **Table 2** shows various social groups and their preferred definitions of metaverse on four main constituent themes.

The self-centred positions amongst different social groups potentially challenge the visions of the metaverse by Parisi and the "Open Metaverse"[85] initiatives by Cesium and Epic Games.

80 G. D. Ritterbusch and M. R. Teichmann, "Defining the Metaverse: A Systematic Literature Review," in IEEE Access, vol. 11, pp. 12368-12377, 2023

81 H. M. Cooper, "Organizing knowledge syntheses: A taxonomy of literature reviews," Knowl. Soc., vol. 1, no. 1, pp. 104–126, Mar. 1988

82 J. P. Girard and J. Girard, "Defining knowledge management: Toward an applied compendium," Online J. Appl. Knowl. Manage., vol. 3, no. 1, pp. 1–20, 2015

83 Dolata, M., & Schwabe, G. (2023). What is the Metaverse and who seeks to define it? Mapping the site of social construction. Journal of Information Technology.

84 https://en.wikipedia.org/wiki/Social_construction_of_technology

85 https://www.buildingtheopenmetaverse.org/

Table 2: Relevant social groups and their preferences/desires on each component of the metaverse (From Dolata and Schwabe, 2023)

Theme	Group	Producers			Users	
		Big tech	Non-commercial providers	Game Producers	Individual users	Retail and entertainment
Ontological	O1-Position	Extending reality.	Extending reality.	Disconnected from reality.	Disconnected from reality.	Mirroring reality. Blended with reality.
	O2-Category	Platform. Development of Internet.	Universe or set of worlds.	Game. Single world.	Place, realm, environment.	Place, realm, environment.
	O3-Status	To come into existence.	Coming into existence.	Existing here and now.	Existing here and now.	Existing here and now.
Differential	Central Difference	Immersive 3D presentation.	Economic activity as a core. There are no specific objectives.	Created by members.	More realistic experience. Persistence.	More data about an individual.
Structural	S1-Central Constituents	Content. Avatars. Places/ Spaces.	NFTs. Virtual Objects. Places/Spaces.	Places/ Spaces. Avatars.	Places/Spaces. Virtual objects.	Places/ Spaces. Virtual objects. Avatars.
	S2-Relationships within the Metaverse	Habitation. Interaction.	Interaction. Authorship. Ownership.	Interaction. Authorship.	Interaction. Authorship.	Interaction. Ownership.
	S3-Relationships with outside Entities	Centralized. Closed. Interoperable.	Decentralized. Open. Interoperable.	Centralized. Closed. Competition.	Decentralized. Open. Interoperable.	Decentralized. Open. Interoperable.
Capabilities	Centrally featured interactions	Participating. Socializing. Collaborating. Working.	Trading. Changing. Entertaining.	Entertaining. Socializing. Changing.	Socializing. Changing. Trading.	Entertaining. Participating. Trading.
	Impact	Improvement.	Improvement.	Improvement.	Improvement.	Improvement.

Advocates			Bystanders	
(Crypto) investors	Affirmative governments	Agnostic governments	Sceptics	Critics
Mirroring reality.	Mirroring reality.	Disconnected from reality.	Any.	Disconnected from reality.
Universe or set of worlds.	Place, realm, environment.	Development of the Internet.	Buzzphrase or hype.	Form of existence.
Coming into existence.	Coming into existence.	To come to existence.	To come into existence.	Existing here and now.
Economic activity as a core.	3D (not 2D) presentation. Not a single technology.	More data about an individual. Economic activity as a core.	Not a sum of technologies. Immersive 3D presentation.	More realistic experience. More data about an individual.
NFTs. Cryptocurrencies. Virtual assets.	Digital twins. Places/Spaces.	Cryptocurrency. Avatars.	Any.	Places/ Spaces. Avatars. Content.
Ownership. Representation.	Representation. Interaction.	Ownership. Interaction. Authorship.	Interaction. Ownership. Habitation.	Habitation. Interaction.
Decentralized. Open. Interoperable.	Centralized. Competition.	Centralized. Closed.	Competition.	Centralized. Closed. Competition.
Trading. Entertaining.	Governing. Teleporting. Changing.	Trading. Socializing. Entertaining.	Doing nothing/ anything.	Entertaining. Socializing. Doingnothing/ anything.
Improvement.	Improvement.	Neutral/Decay.	Neutral.	Decay.

Themes

Ontological:

What kind of thing is the metaverse? (O1); Does the metaverse exist at all? (O2); How does the metaverse exist? (O3)

Differential:

What the metaverse is and not? How does it differ from other phenomena, including video gaming and social media?

Structural:

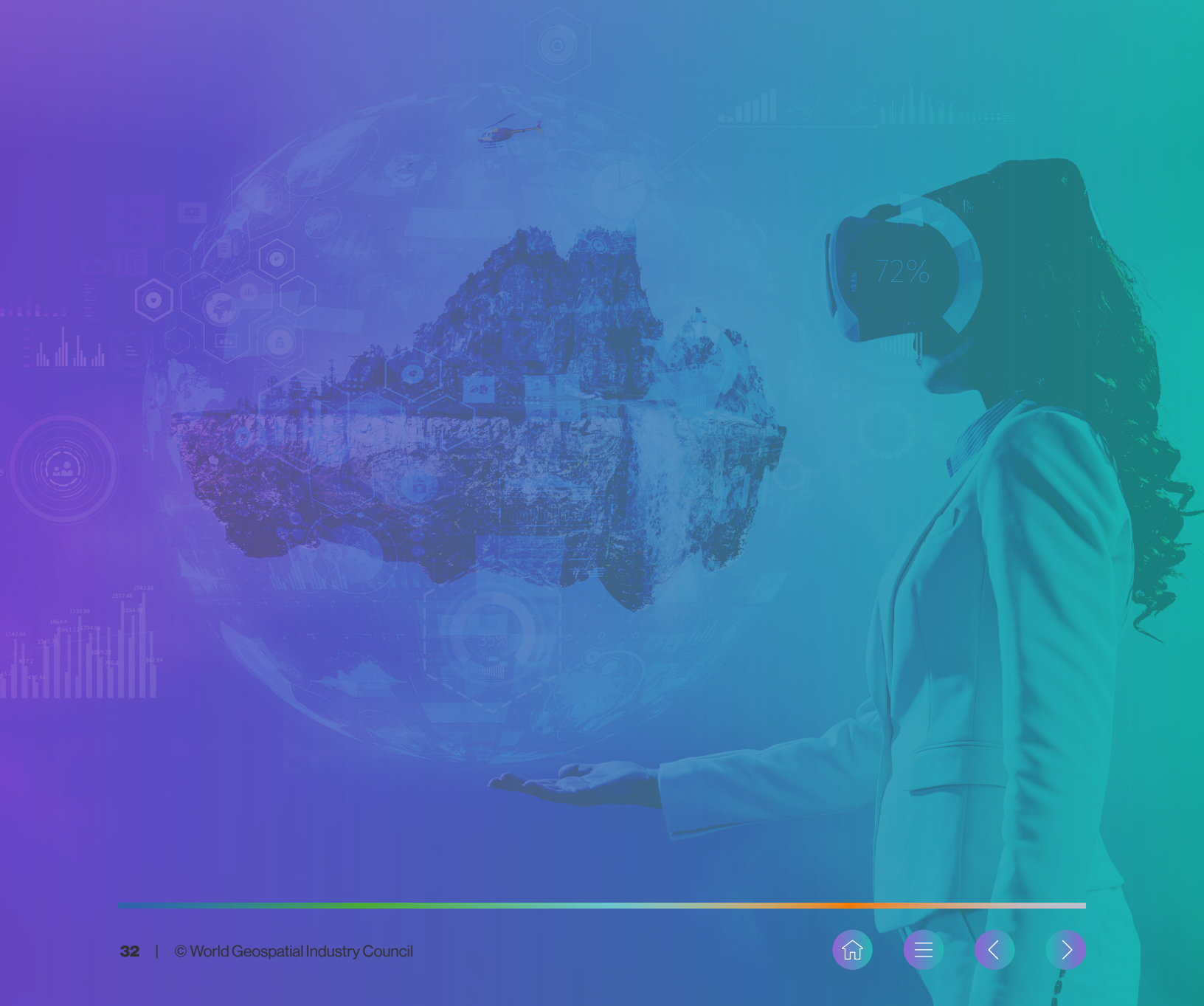
What are the entities in the metaverse (S1), and how do these entities inside the metaverse relate to each other (S2) and to the entities outside the metaverse (S3)?

Capabilities:

What are the capabilities and use cases of metaverse?

Appendix 3

Bringing Geospatial Context to the Metaverse – Experiences of WGIC Member Companies



Esri's flagship ArcGIS suite of products, namely ArcGIS Pro, ArcGIS Enterprise, and ArcGIS Online, along with the respective supporting sub-component products, have demonstrated their capability to create, visualize, publish, and manage content and services for both 2D and 3D data in online digital environments. ArcGIS products provide sophisticated geospatial analysis and administrative capabilities such as case management with its rich suite of built-in and software development kits. Esri products have been used in numerous geospatial applications in all working environments, ranging from academics to environment to defense and AEC. Esri has recently strengthened its software capabilities for real-time data feeds from IoT sensors and indoor mapping. Combined with their public outreach, Esri products can be seen as enablers in almost all the layers of the metaverse value chain.

The following list is only a small selection of Esri products that can be used to create various aspects of digital twins.

3D Modelling Products: ArcGIS Reality can be used to create workflows to convert drone and aerial imagery of an area of any size into survey-grade 3D models rapidly. It is available offline (e.g., in the field) or as a cloud service. *ArcGIS CityEngine* is a 3D modeling software for creating massive, interactive, and immersive urban environments that can be either based on fictional worlds (procedural modeling) or real-world geospatial data. One of the unique aspects of the product is its integration with several prominent gaming engines (e.g., Unreal, Unity), widely used 3D modeling applications in the computer graphics industry (e.g., 3Ds Max, Maya, Rhino, Houdini), and metaverse platforms (e.g., NVIDIA Omniverse)



Figure 8: Visualization of a procedural 3D model in ArcGIS CityEngine (Source: Esri)

Built and General Infrastructure Modelling and Management: *ArcGIS Indoors* can create new and combine existing 2D/3D representations of interior spaces into a structured single ArcGIS Indoors Information data model, i.e., a BIM/digital twins. *ArcGIS GeoBIM* can connect data, process, and task management from geospatial and BIM systems in a single user interface, allowing greater situational awareness and seamless collaboration across different teams. *ArcGIS Urban* can be used to create, track, and review development projects with a digital twin of a city. It provides functions to simulate what-if scenarios and collaborative decision-making.



Figure 9: Esri products to create and integrate datasets for urban digital twins, and manage projects from site-level to entire city level (Source: Esri)

Real-time Data: *ArcGIS Velocity* is a cloud-native application that enables users to ingest real-time data feeds from IoT platforms, message brokers, or third-party APIs. It can also process, visualize, and analyze real-time data feeds, store them as big data, and perform fast queries and analysis.

Indexed 3D Scene Layers (I3S), developed by Esri and contributed to OGC, is a community standard for streaming and storing massive amounts of geospatial content. I3S has been rapidly evolving to capture new use cases and techniques to advance geospatial visualization and analysis. The current version, OGC I3S 1.3 enables efficient transmission of various 3D geospatial data types, including discrete 3D objects with attributes, integrated surface meshes, and point cloud data covering vast geographic areas as well as highly detailed BIM content to web browsers, mobile apps, and desktop.



Figure10: Use of ArcGIS Velocity to monitor the traffic flow patterns in New York City buses (Source: Esri)

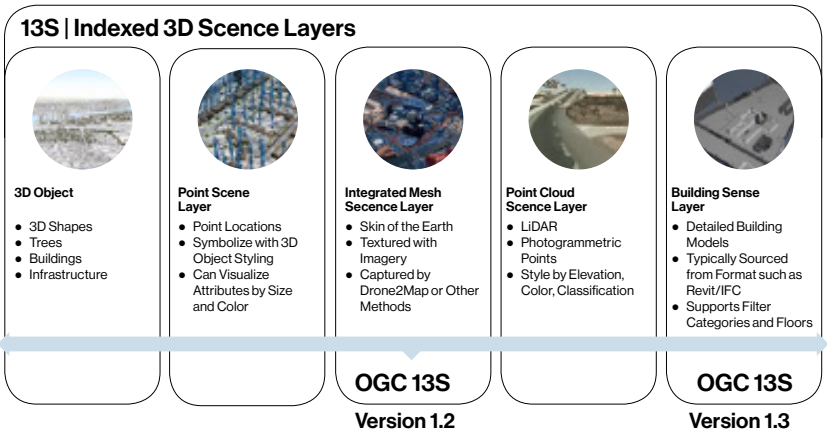


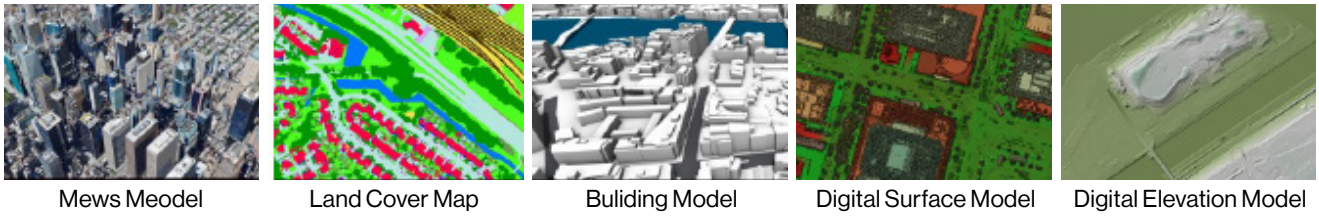
Figure 11: Different types of layers supported in OGC I3S Community Standard (Source: Esri)

For more information visit
https://www.esri.com/en-us/arcgis/products/index.

HEXAGON

Hexagon AB products and services cover the entire range of components in the geospatial ecosystem. They are the enablers of the creator economy, spatial computing, and infrastructure layers in the metaverse value chain. This conglomerate provides complete stack solutions that combine sensors, software, autonomous technologies, and platforms to serve the solutions.

MetroHD, in the HxGN Content Program, is a suite of ultra-high-definition 2D and 3D geospatial vector and raster data sets derived from 5 cm resolution imagery and 20 points/m² density LiDAR data, which are used in the creation of the virtual models of structures in urban digital twins (**Figure 12**).



Mews Meodel | Land Cover Map | Buliding Model | Digital Surface Model | Digital Elevation Model

LiDAR Point Cloud | Oblique Imagery | True Orthophoto | Standadr Orthophoto | Tree Model

Figure 12: MetroHDR product set (Source: Hexagon)

Hexagon Digital Reality (HxDR) (**Figure 13**) and **LuciadRIA** (**Figure 14**) platforms provide a rich set of tools to share, store, analyze, process, and manage the virtual models in digital twins, including real-time sensor data.

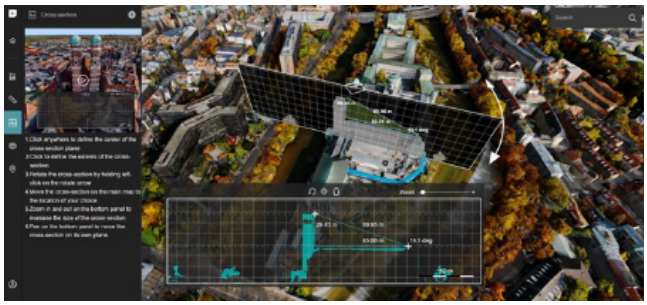


Figure 13: Figure 13: HxDR (Source: Hexagon)

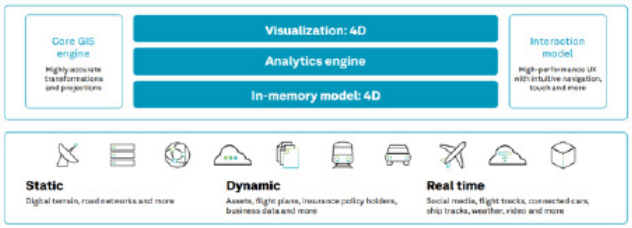


Figure 14: Figure 14: LucidRIA (Source: Hexagon)

For more information visit
https://hexagon.com/industries/mapping-geospatial-content

ORACLE

Oracle products and its experts specialize in their proprietary Oracle Database software and Oracle Cloud Services. A noteworthy free component of the Oracle Database, Oracle Spatial and Graph, enables the management and deployment of geospatial data in a native type within the Oracle Database. These offerings are used in precision agriculture, smart buildings, smart cities, digital twins, smart meters, and distributed generation in utilities. Oracle software, hardware, and expertise technology will be foundational in enabling the creator economy", spatial computing, decentralization, and infrastructure layers of the metaverse value chain.

Oracle Smart Construction Platform (OSCP) is a suite of interconnected desktop and cloud-hosted applications designed to handle all types of data and information required to create and utilize assets and their digital twins. The applications cover the entire lifecycle of projects. While virtual reality is often, and perhaps unhelpfully, portrayed as a central driver of the metaverse, technologies such as OSCP will provide a unique value proposition and workhorse for making it part of daily lives.

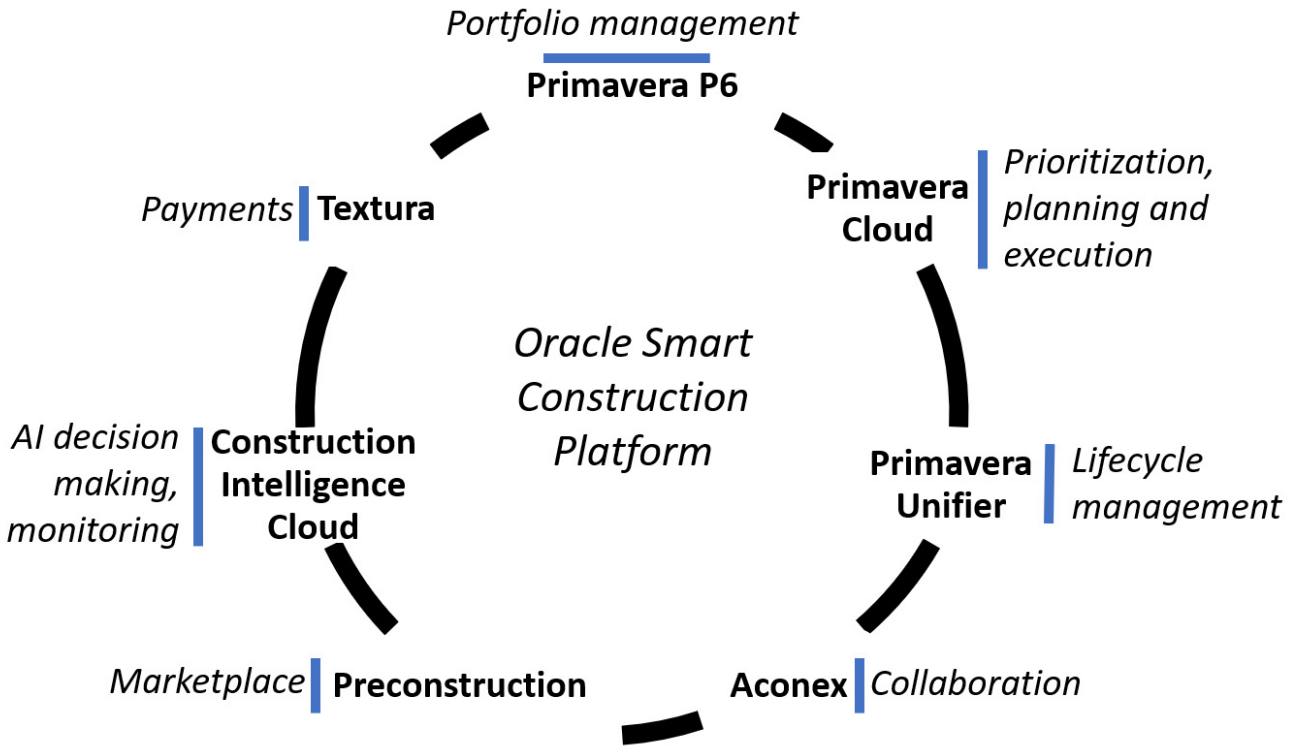


Figure 15: Components of the Oracle Smart Construction Platform

Oracle Industry Innovations Labs (OIIL) spread worldwide provides customers a hands-on environment to develop new ideas and create solutions leveraging technology from Oracle. With industry partners, the Oracle team at the Dry Dock in the Chicago site of the OIIL has developed reality mapping solutions that take reality capture data and transform it into 2D and 3D reality maps of indoor and outdoor project environments. These experiments have provided valuable insights for creating BIM and digital twins for metaverse, such as:

- ✔ Autonomous capture and processing of reality using self-flying or self-operated drones is practically possible,
- ✔ As expected, high-quality LiDAR and imaging solutions outperform smartphones in geometrical accuracy and completeness of the outputs; however, the portability, pace, and ease of capturing using a smartphone was uniquely valuable for reality capture in confined spaces (e.g., underground utilities, stairwell, electrical systems)

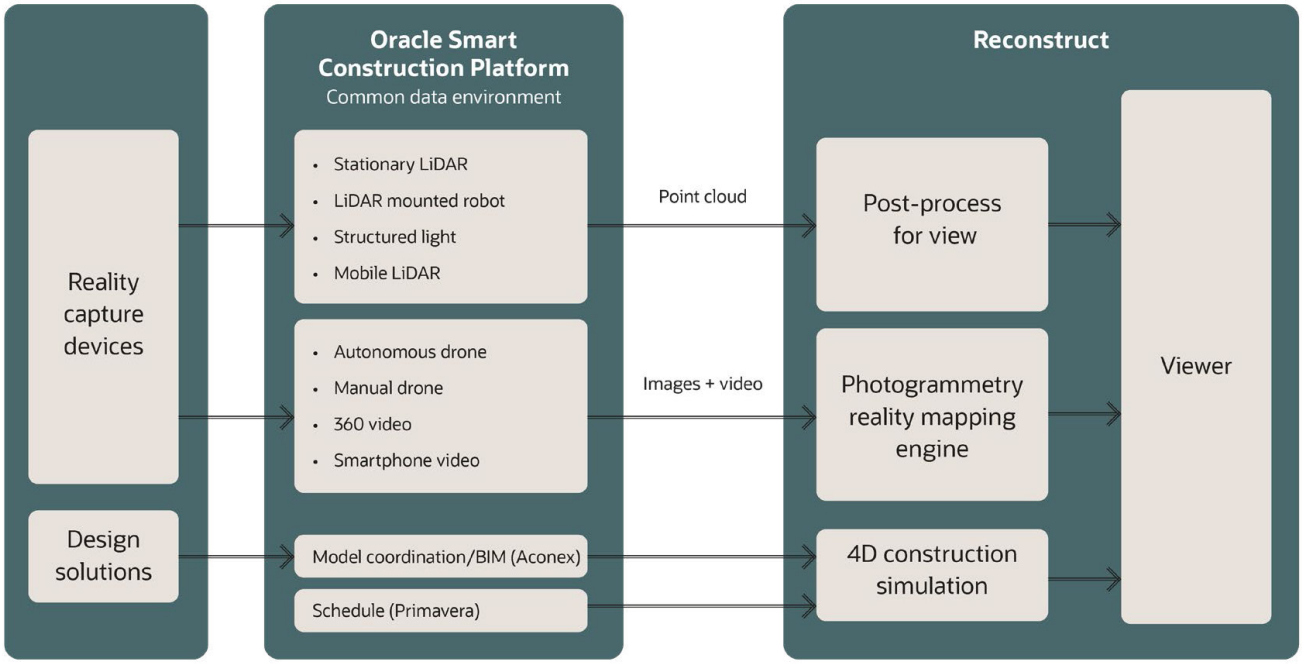


Figure 16: Data import process and the workflow for transforming reality capture into actionable and exportable results (Source: Oracle)

To understand the metaverse's system integration aspects, OILL has built a Connected Hub **(Figure 17)** This interactive, innovative town model details energy supplies, intelligent meters, and vehicle charging. This can simulate scenarios such as outages, deploy field crews, maintain assets, and customer experience.



Figure 17: Connected Hub model (Source: Oracle)

For more information visit
https://www.oracle.com/

Schneider Digital is primarily a hardware supplier in the geospatial ecosystem, focusing on stereoscopic display systems (desktop and large-format). Schneider Digital products are enablers of the creator economy and spatial computing layers in the metaverse value chain.

PluralView consists of stereoscopic desktop displays combined with lightweight and passive polarization glasses, enabling fatigue-free, stereoscopic work throughout the day. The application scenarios could range from creating new 3D content (**Figure 18**) to conventional casework-type 3D data visualization scenarios (**Figure 19**).

Schneider Digital offers two solutions for a more collaborative environment using 3D Data. **Smart VR Wall (Figure 20)** is a large-format (up to 12 sq. m. at 6K resolution), front-projection stereoscopic display wall with active LCD shutter glasses. **3D GlobeView (Figure 21)** consists of stereoscopic, single-panel monitors at diagonal sizes between 79" and 86 native 4K resolution and employs a line-interlaced stereoscopic display principle for viewing with passive, polarized glasses by groups of up to 20 people.

Schneider Digital continues expanding its hardware and software portfolio of stereoscopic visualization technology by focusing on capabilities to generate a stereoscopic output on any stereo visualization device (including VR and AR headsets). In particular, they are developing a competent **Stereo Mouse Cursor** (SMC). SMC will enable advanced and user-friendly 3D object navigation and editing control across all visualization platforms. The SMC can be compared to a laser pointer, leaving a dot or cross-hair on the designated object, enabling associated view control and object editing capabilities in the metaverse and standard 2D displays.

For more information visit
https://www.schneider-digital.com/en/



Figure 18: Content Creation in PluralView



Figure 19: Casework management in PluralView



Figure 20: SmartVR Wall



Figure 21: 3D Globeview

(Source: Schneider Digital)

TomTom delivers high-accuracy maps of the world accessible on various display and computing form factors. TomTom's expertise will be essential in the creator economy and spatial computing layers in the metaverse value chain.

Mobile mapping (MoMa) -TomTom is an expert in intelligent mapmaking, a highly efficient hybrid approach that combines traditional mapmaking practices with more advanced ones, such as probe and sensor data, to collect data at two-centimeter accuracy.

MoMa vehicles have the latest 360 Ladybug cameras and Velodyne LiDAR. Combined with DGPS (Differential GPS), these are used to collect more than 700,000 data points to an accuracy within two centimeters.

The equipment captures the road surface and furniture, such as signs, lights, and barriers. Approximately 3.8 billion pixels are captured for every kilometer driven, resulting in high-quality images.

TomTom collects more than three million kilometers of data each year and has a repository of more than ten million kilometers of up-to-date images from which customers can commence feature extraction immediately. TomTom has a global fleet of active MoMa vehicles that can support customer requests in multiple geographies (**Figure 22**).



Figure 22: : Large standardized archive in all key markets (Source: TomTom)

Features

- ✔ High-definition feature-rich images and point cloud sensor data
- ✔ MoMa Application Programming Interface (API) published on developer.tomtom.com
- ✔ Supports licensing of already collected MoMa data (archives); as an exception (case-by-case decision),
- ✔ TomTom can support on-demand collection to close specific minor coverage gaps in areas with confirmed value for TomTom

Benefits

- ✔ Delivers high-quality images - 3.8 billion pixels per kilometer - with good clarity to enable features to be extracted
- ✔ Provides controlled access to MoMa imagery and point cloud sensor data
- ✔ Enables instant access to the existing MoMa data, as well as the opportunity to request access to new data through on-demand collection
- ✔ Ensures there is a solution for closing potential coverage gaps

Sample Applications

The map data extracted by TomTom's MoMa-as-a-service is designed to be integrated with a customer's business applications and asset management tools. The primary use is feature extraction; almost any feature captured within an image can be extracted for analysis or to create a database of assets.

Such features include:

- ✔ Street lighting
- ✔ Road signage
- ✔ Street furniture such as seating, shelters, or barriers
- ✔ Road markings
- ✔ Bridge types and heights

MoMa is not a technology that a local authority, government, or individual business would typically invest in, as the costs of equipping and maintaining a fleet would be high for limited use cases. This makes TomTom the ideal service provider.

For more information visit
http://www.tomtom.com



Figure 23: Use of MoMa data in spatial digital twins (Source: TomTom)

Trimble produces software, data, hardware, and training for survey-grade reality mapping, near-real-time positioning, and all aspects of project management. These software workflow contexts could range from the initial design of the 3D models of infrastructure to an AR/MR visualization on site, fabrication, and daily project management. These systems and solutions enable the metaverse value chain's creation economy, spatial computing, human interface, and infrastructure layers

Built and General Infrastructure Modelling and Management: *Construction One* is a construction management solution suite with 40 seamlessly connected applications covering every project phase, demonstrating interoperability and integrations among several application domains. The applications range from data gathering in the field to construction/fabrication, financial management, and project management.

Mixed Reality: Connect AR, XR10 with Holo Lens2, and FiedLink MR provide interactive visualization of true-to-life scale overlays of 3D data onsite in the field. These also have applications to share, communicate, and manage tasks with colleagues.

For more information visit
https://www.trimble.com/



Figure 24: The use of Trimble Construction One to deliver a construction project from design stage to building, and delivery (Source: Trimble)



Figure 25: Onsite use of Trimble augmented, mixed, and extended reality solutions using Microsoft HoloLens 2 (Source: Trimble)

Appendix 4

OVERTURE MAPS FOUNDATION

Founded in 2022 by Amazon Web Services, Meta, Microsoft, and TomTom, OMF is a) sourcing, curating, and publishing high-quality, up-to-date global 2D and 3D (buildings) map data quality; and b) developing a Map Schema (e.g., vocabulary) and Global Entity Reference System (GERS) to make datasets from disparate sources interoperable.

OMF datasets will be openly available and licensed under the Community Database License Agreement – Permissive v2 (CDLA). However, if the source datasets were published under a different license, these will inherit the source data license. For example, data derived from OpenStreetMap (OSM) will be released under the Open Data Commons Open Database License (ODbL v1.0) because OSM uses ODbL1.0.



Figure 26: Excerpt from the OMF map (Source: OMF)

The OMF is noteworthy because global, open, and interoperable geospatial datasets will be fundamental to keeping the metaverse seamless and inclusive.

The first release **(Figure 26)** of worldwide OMF Map data has four themes, namely,

- ✔ **Places of interest (POI) –** There are over 59 million POI records contributed to the OMF by Meta and Microsoft.
- ✔ **Buildings –** These are 780 million unique building footprints worldwide based on a combination of various open data from OSM, Microsoft AI-generated building footprints, and Esri.
- ✔ **Transportation –** This is a worldwide road network derived from OSM and structured to allow additions of speed limits and real-time traffic.
- ✔ **Administrative Boundaries –** This global open national and regional administrative boundaries dataset. It includes regional names translated into over 40 languages to support international use, i.e., interoperability in different local systems.

For more information visit
https://overturemaps.org/





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https://wgicouncil.org/

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